

Enhancing Pedestrian Detection in Low-Light Environments with RGB-IR Fusion

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Abstract

Pedestrian detection in low-light conditions remains challenging due to the limitations of RGB cameras in dark environments. While infrared (IR) imaging can highlight thermal targets, it lacks the structural and textural details of visible imagery. To address this, we propose a novel attention-guided dual-stream fusion network that generates 3-channel RGB-like images by combining modality-specific features from IR and RGB inputs. Our model incorporates a dynamic IR modulation block and a multi-component loss function designed to preserve semantic alignment, structural consistency, and perceptual fidelity. The fused outputs are directly compatible with pretrained RGB object detectors. When evaluated on the KAIST dataset, our pipeline—using fused inputs with a Deformable DETR model—achieves an mAP@0.5 of 0.9088, compared to 0.5500 for RGB-only inputs. This represents a 35.9 percentage point absolute gain and a 65.2% relative improvement. Code, pretrained models and video are available at <https://drive.google.com/drive/folders/11nOg-YSqk2eIU98jUsBUi7ysdos1A376?usp=sharing>.

1. Introduction

Pedestrian detection is a critical component of modern intelligent transportation and surveillance systems, enabling applications such as autonomous driving, smart city infrastructure, and public safety monitoring. While current state-of-the-art detectors achieve strong performance under ideal lighting conditions, their effectiveness deteriorates drastically in nighttime scenarios. This is largely due to the limitations of standard RGB imagery, which loses detail and contrast in low-light environments—precisely when robust detection is most vital.

To address this challenge, thermal infrared (IR) imagery is often introduced as a complementary modality. IR sensors capture thermal radiation emitted by objects, making pedestrians visible even in total darkness. However, IR im-

ages lack fine semantic details and color information, making them insufficient alone. This motivates the use of RGB-IR fusion, combining the semantic richness of RGB with the thermal robustness of IR to create a more informative representation.

Despite extensive research in multispectral fusion, most approaches perform fusion at the feature level within the detection model, requiring architecture modifications and often lacking interpretability. Moreover, the resulting fusion may not be directly compatible with pretrained RGB detectors, limiting their practicality.

In this project, we propose an attention-guided dual-stream fusion model that learns to generate a 3-channel RGB-like image from paired RGB and IR inputs. This fused image preserves the structure and thermal cues of IR while enhancing visual fidelity with RGB features. The goal is to leverage this fused output for robust nighttime pedestrian detection using a transformer-based object detector.

To that end, we employ Deformable DETR, a transformer-based detection architecture that is particularly well-suited for this task due to its flexible attention mechanism and capacity to adapt to varying input distributions. Its sparse, deformable attention allows it to focus on key spatial regions of interest, making it effective at handling the structural diversity introduced by fused RGB-IR content. To facilitate this adaptation, we follow a two-phase fine-tuning strategy. This staged process enables the model to bridge the domain gap between natural RGB and fused imagery, ultimately improving detection performance in low-light conditions.

2. Related Work

Early methods for infrared-visible (IR-VIS) image fusion relied on multi-scale signal processing techniques, such as pyramid and wavelet transforms, to combine thermal and visible data. The advent of deep learning brought significant improvements, introducing end-to-end architectures that better preserve both thermal highlights and fine textures.

DenseFuse (Li & Wu, 2019) proposed a densely connected encoder-decoder CNN with two fusion strategies, outperforming prior models in subjective and objective evalu-

ations. IFCNN (Zhang et al., 2020) offered a general-purpose CNN framework adaptable to multiple fusion modalities, including IR-VIS. NestFuse (Li et al., 2020) improved upon these models by introducing spatial- and channel-wise attention within nested skip connections, enhancing multi-scale feature interaction. More recently, CrossFuse (Xu et al., 2022) incorporated dynamic kernels and cross-scale feature alignment to merge modality-specific representations more effectively, especially under diverse lighting conditions.

Building on these architectures, attention-guided and dual-stream fusion networks have become increasingly prominent. These models typically use separate encoders for visible and infrared modalities and fuse their features using attention mechanisms. For example, SCGRFuse and EMA-Fusion (Zhou et al., 2021) used Convolutional Block Attention Modules (CBAM) to dynamically reweight modality contributions, while cross-modal attention has been applied to improve feature alignment and interpretability.

In the domain of pedestrian detection, particularly under low-light conditions, datasets such as KAIST (Hwang et al., 2015) and LLVIP (Jia et al., 2021) have driven the development of multispectral detection systems. Early multispectral detectors adapted existing architectures (e.g., Faster R-CNN) to support dual-input streams and explored fusion at various stages (early, halfway, or late), with halfway fusion found to be most effective (König et al., 2017). Illumination-aware models such as IAF R-CNN (Li et al., 2019) and MCFF-YOLOv4 (Liu et al., 2021) introduced dynamic weighting schemes to adapt fusion based on scene brightness.

Transformer-based detectors like MS-DETR (Xing et al., 2023) extend DETR to multispectral inputs using dual CNN backbones and a cross-modal transformer decoder with adaptive attention. Similarly, hybrid models such as DAMSDet (Althoupety et al., 2024) combine CNN encoders with transformer decoders to capture long-range dependencies. However, these approaches often require specialized architectural modifications and cannot leverage pretrained RGB weights, limiting their plug-and-play applicability.

Motivated by recent attention-guided dual-stream designs, we propose a fusion pipeline explicitly optimized for nighttime pedestrian detection. Our architecture introduces a novel infrared (IR) modulation block that dynamically reweights thermal features based on local contrast and illumination. This mechanism allows the network to emphasize IR cues in dark regions while preserving RGB details in well-lit areas. In addition, we design an adaptive multi-component loss that guides the network to produce perceptually coherent and structurally faithful outputs. This loss integrates brightness-aware RGB reconstruction, gra-

dient preservation using Sobel filters, cosine-based IR feature alignment, VGG16-based perceptual similarity, SSIM, and artifact suppression. Together, these components enable the model to learn context-sensitive fusion mappings. Crucially, our system outputs a 3-channel RGB-like image, allowing seamless integration with pretrained detectors like Deformable DETR—without modifying their architecture. This ensures compatibility with standard pipelines and facilitates effective pedestrian detection in extreme low-light conditions.

3. Problem Definition

Nighttime pedestrian detection poses a significant technical challenge due to the degradation of input data quality under poor illumination. RGB sensors, which rely on visible light, suffer from low contrast, loss of texture, and high sensor noise in dark environments. This leads to unreliable feature extraction in convolutional backbones and degraded region proposals or classification confidence in object detectors. Pedestrians wearing dark clothing or partially occluded in shadows are often misclassified or missed entirely. The features extracted from nighttime RGB images often become spatially ambiguous or semantically weak, impairing the attention mechanisms in modern detectors.

While infrared (IR) imagery provides strong object-level contrast by capturing thermal emissions, it lacks semantic richness and fine-grained detail (e.g., clothing texture or contextual background information), which are essential for accurate localization and classification. Simply concatenating RGB and IR channels or performing naive feature fusion introduces *modality conflicts*—where the two domains exhibit different noise distributions, resolutions, and content density—which can confuse feature extractors and degrade learning. This is particularly problematic in transformer-based models that rely heavily on feature attention across tokens, where misaligned or irrelevant modality signals can disrupt attention flow.

The key problem, then, is how to create a fused representation that:

- Preserves the thermal structure and contrast of IR for robust pedestrian localization;
- Retains semantic context and appearance from RGB for fine-grained classification;
- Is structurally and visually compatible with RGB-pretrained detection pipelines.

Moreover, existing pretrained detectors such as Deformable DETR are trained exclusively on natural RGB images and cannot be expected to generalize to fused or synthetic inputs without careful domain adaptation.

These limitations highlight the need for a robust representation that can unify the complementary strengths of RGB and IR while mitigating their individual weaknesses. Such a representation must be able to maintain the spatial structure and thermal contrast from IR, preserve the semantic content from RGB, and remain compatible with existing detection architectures. Furthermore, any solution must address the domain shift between natural RGB data and fused imagery, ensuring that modern detectors can still generalize effectively without retraining from scratch.

This motivates the development of fusion strategies that are not only modality-aware but also model-aware—capable of producing detector-friendly outputs that retain interpretability and detection performance in challenging nighttime scenarios.

4. Methodology

4.1. Dataset and Preprocessing

We use the KAIST Multispectral Pedestrian Dataset, which provides paired RGB and thermal infrared (IR) images captured in both day and night conditions. Our experiments focus on nighttime images, with a small portion of daytime samples included for generalization.

To normalize the data, we compute the mean and standard deviation for each modality using 500 randomly sampled training images. RGB images are normalized channel-wise after scaling to $[0, 1]$, while single-channel IR images are normalized globally. All inputs are resized to 256×256 to ensure consistent dimensions.

Normalization is applied as follows:

$$\text{rgb}_{\text{norm}} = \frac{\text{rgb}/255.0 - \mu_{\text{rgb}}}{\sigma_{\text{rgb}}}, \quad \text{ir}_{\text{norm}} = \frac{\text{ir}/255.0 - \mu_{\text{ir}}}{\sigma_{\text{ir}}}$$

Images are converted to PyTorch tensors and reshaped to $[3, H, W]$ for RGB and $[1, H, W]$ for IR. Each processed sample is stored as a `.pt` file containing the normalized tensors, original file name, and dataset statistics. This preprocessing ensures reproducibility and stable model training.

4.2. Fusion Model Architecture

Our fusion model follows a dual-stream encoder-decoder architecture with attention-guided feature modulation and skip connections. It receives paired RGB and IR images as input and outputs a 3-channel RGB-like fused image.

As shown in Figure 1, the RGB and IR inputs are encoded independently through two convolutional encoder branches. These feature maps are then fused using dual-path attention modules and IR-specific modulation. The

resulting fused representation is decoded through transposed convolutions, with skip connections from both encoder paths to preserve low-level details. A tone adjustment module refines the final output.

The model output is normalized in the range $[-1, 1]$, preserving both thermal and visual cues for effective detection.

4.3. Fusion Loss

A multi-component loss function was designed to preserve both the structural and semantic integrity of RGB and IR modalities, particularly under low-light conditions.

Reconstruction Loss: An adaptive L1 loss is computed between the fused output and RGB input, weighted by brightness masks that emphasize high-frequency details in bright regions while deemphasizing underexposed areas.

Gradient Preservation: Sobel-based gradient magnitudes are computed on both the fused image and IR input to preserve thermal edges. This is applied at full and half resolutions, with weights increasing in darker regions to reinforce contour fidelity in low-light scenes.

IR Feature Alignment: A cosine similarity loss is computed between the fused green channel and IR input at dark pixels, encouraging semantic alignment in thermal-salient regions.

Perceptual and Structural Losses: A VGG16-based perceptual loss is computed between normalized fused and RGB images to maintain natural visual features. Structural similarity (SSIM) loss further reinforces global structure.

Artifact Suppression: To reduce visual noise, we include losses that penalize checkerboard artifacts (via L1 difference in interleaved pixels) and color inconsistencies (via channel-wise color deviation).

The total loss is a weighted sum of these components. Additionally, the model adaptively lowers its dark region threshold over time to progressively focus training on challenging low-light conditions.

The fusion model is trained using Adam optimizer with ReduceLROnPlateau scheduling and the multi-component loss. Training monitors total loss and dark-region ratio; the best model is saved based on validation loss. Once training converges, our model produces fused 3-channel images that incorporate both IR and visible information. It was trained for 15 epochs using a batch size of 15 and a learning rate of $1e-4$.

5. Fine-Tuning Deformable DETR

After generating fused RGB-like images with our fusion model, we fine-tune a Deformable DETR model for pedes-

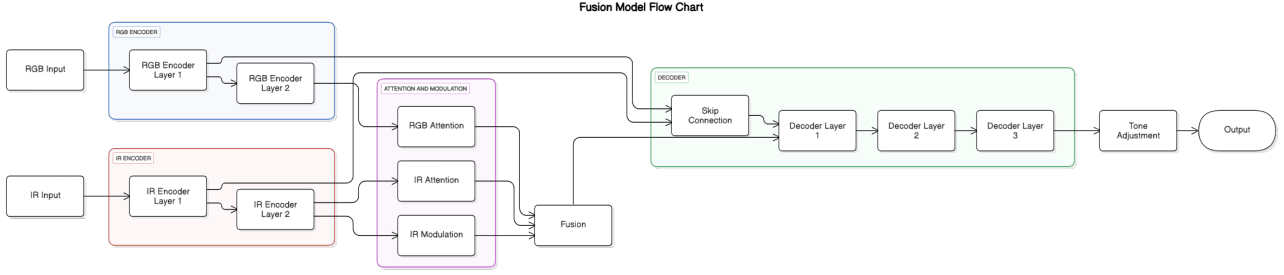


Figure 1. Fusion model flow chart illustrating dual-stream encoding, attention-guided fusion, skip connections, and decoder with tone adjustment.

trian detection under low-light conditions. To adapt effectively while preserving pretrained knowledge, we employ a two-phase fine-tuning strategy with layer-specific learning rates.

5.1. Layer-Specific Learning Rates

Model parameters are grouped by functional role (e.g., encoder, decoder, detection head) and assigned distinct learning rates. Higher learning rates are allocated to the detection head and decoder to accelerate adaptation, while earlier layers receive smaller learning rates or remain frozen. Only parameters with non-zero learning rates are passed to the optimizer, ensuring controlled fine-tuning.

5.2. Phase-Aware Freezing Strategy

To mitigate catastrophic forgetting, we adopt a staged unfreezing schedule: Phase 1 unfreezes only the detection head, decoder, and layer 4 of the ResNet backbone; all other layers remain frozen. Phase 2 additionally unfreezes the Transformer encoder and layer 3 of the backbone, enabling gradual adaptation of deeper features using smaller learning rates.

5.3. Training Strategy and Optimization

Training is performed in mixed precision with gradient scaling and clipping for stability. We use a warmup + linear decay learning rate scheduler in Phase 1, transitioning to cosine decay in Phase 2. Each epoch includes standard supervised training with validation based on loss and mAP@0.5. Early stopping is applied if validation performance plateaus.

5.4. Baseline Comparison

To assess the benefit of fusion, we also finetune another Deformable DETR model on nighttime RGB images only. This provides a baseline for evaluating the effectiveness of fused inputs under low-light conditions. Lighter finetuning was used for the RGB-based model (decoder + heads only,



Figure 2. Fusion results on a nighttime scene.

10 epochs) given that Deformable DETR is pretrained on RGB COCO. In contrast, the fused model. However, the architecture is held constant and the same data split was used.

6. Experimental Results

6.0.1. FUSION RESULTS

Figure 2 showcases the output of our fusion model under nighttime conditions. The RGB input (left) suffers from low visibility in shadowed regions, making pedestrian detection challenging. In contrast, the IR input (center) captures thermal signatures, clearly highlighting pedestrians but lacking semantic and color information. The fused output (right) effectively combines both modalities—retaining the semantic richness and structure of the RGB image while enhancing visibility through thermal cues. This balanced representation significantly improves interpretability and detection robustness in low-light scenarios.

Metric	Score
Average SSIM	0.3843
Average PSNR	8.58 dB
IR Gradient Preservation	0.9556
Mutual Information (IR ↔ Fused)	1.4941

Table 1. Fusion quality metrics evaluated on the KAIST nighttime validation set.

6.0.2. DEFORMABLE DETR FINETUNING RESULTS

To evaluate the effectiveness of our fusion model for pedestrian detection under low-light conditions, we fine-tuned a Deformable DETR detector on two input modalities: (1) RGB-only nighttime images and (2) 3-channel fused images generated by our attention-guided fusion network. Table 2 presents the detection performance in terms of mAP@0.50 on the KAIST nighttime validation set.

Table 2. Detection performance of Deformable DETR fine-tuned on RGB-only vs. fused inputs.

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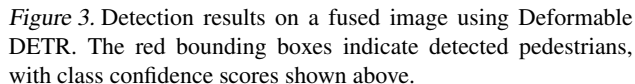


Figure 3 illustrates qualitative results from our pipeline, where a Deformable DETR detector is applied to a fused 3-channel image generated by our attention-guided fusion network. The original scene—captured under extreme low-light—poses a challenge for RGB-only detectors due to underexposure and low contrast. However, the fused image retains sharp thermal contours of the pedestrians from the IR input, while restoring contextual scene features from the RGB modality.

7. Contributions

- We introduce an attention-guided dual-stream fusion network with a modality-specific infrared modulation block that dynamically adjusts the influence of thermal features based on local brightness. This improves the preservation of pedestrian silhouettes in low-light

regions.

- An adaptive multi-component loss is proposed, combining brightness-aware RGB reconstruction, gradient preservation, cosine similarity for IR feature alignment, VGG16-based perceptual similarity, structural similarity (SSIM), and artifact suppression. This loss enables perceptually coherent and structurally consistent fusion outputs.
- The proposed fusion network outputs standard 3-channel RGB-like images, enabling plug-and-play compatibility with pretrained RGB detectors such as Deformable DETR, without requiring any architectural changes to the detection pipeline.
- Extensive experiments on the KAIST dataset demonstrate that our method improves mAP@0.5 from 0.5500 (RGB-only) to 0.9088 using fused inputs. This corresponds to a 35.9 percentage point increase and a 65.2% relative improvement in nighttime pedestrian detection performance.

LLM Acknowledgement

This work used the assistance of a large language model (ChatGPT by OpenAI) to improve writing clarity, summarize related literature, and revise LaTeX formatting. Additionally, the LLM was used to refine the syntax of Python code, particularly for preprocessing, model implementation, and visualization. All experimental design, model architecture, training procedures, and analysis were independently developed and executed by the author. The LLM was not used to generate novel content, create experiments, or fabricate results.

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